Simulated-proprioception: from monkey see monkey do to monkey think robot do.

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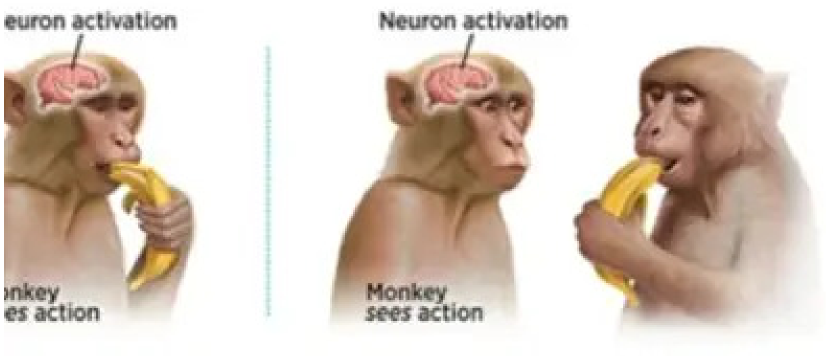
Abstract

Simproprioception is a proof-of-concept brain-computer interface (BCI) pipeline that decodes human motor intent from non-invasive EEG signals and translates those signals into commands for a simulated robot arm. This paper examines the data acquisition bottleneck inherent to Learning from Demonstration (LfD) approaches in robotics, and proposes neuroimaging data as a scalable alternative to expert demonstration. We demonstrate that clinically sourced EEG motor imagery data can serve as a viable input modality for robot command generation, bypassing the expert dependency that constrains conventional LfD pipelines.



Introduction

The Monkey See, Monkey Do Problem

Classical LFD (learning-from-demonstration) approaches work on a simple principle: watch an expert perform a task, record what they did, and train a model to copy it. Monkey see, monkey do.

This framework hides a serious practical problem. Generating training data requires a physically capable expert who can perform the task, that expert to show up repeatedly in a controlled environment, clean recordings of their physical demonstrations, and labels attached to every trajectory they produce. Scale this to real-world robotics and the bottleneck becomes obvious; expert time is expensive, physical demonstrations are difficult to replicate consistently, and the entire data generation process is contingent on human availability and cooperation. The result is a paradigm where half the research lifecycle is spent sourcing data rather than advancing the science.

Monkey Think

This project proposes a different framing. Instead of capturing what a person does, capture what a person intends. The human subject imagines moving their hand. Their motor cortex fires. That neural activity is recorded as EEG, decoded by a trained classifier, and translated into a robot command without the human ever touching the robot. The data generation problem shifts from finding monkeys who can do, to finding data from monkeys who can think. Which is significantly easier.

The population of people who undergo neuroimaging procedures such as EEG, MEG, fMRI, and MRI vastly outnumbers the pool of experts trained to generate LfD-compatible demonstrations. Unlike kinesthetic or teleoperation data, which requires custom collection pipelines built around cooperative, physically capable subjects, clinical neural data is already being collected at scale for medical purposes. The infrastructure for acquiring and processing it is mature, standardized, and widely accessible. The data generation problem does not disappear; it simply moves to a domain where it is far more manageable.

Data availability

Data availability represents one of the most significant bottlenecks in the continued development of robotic systems trained through LfD approaches. Unlike algorithmic limitations, which have seen rapid progress, the data collection process itself remains a persistent structural challenge presenting issues of data redundancy, distributional imbalance, and demonstration inconsistency that compound as task complexity scales [1]. Each demonstration demands significant human effort in the form of expert operation and precise annotation, making it economically infeasible to scale data quantity for the breadth of environmental coverage real-world robotics requires [2]. This challenge is further compounded by a fundamental asymmetry between robotics and other data-driven fields. As Mirchandani et al. note, this aspect of robot data collection drastically differs from modalities such as text or images, where large volumes of data are organically produced by people in their daily activities and are readily available on the web [3]. No such organic pipeline exists for robotics; expert demonstration data must be deliberately constructed, at significant cost, for every new task and environment. Simproprioception proposes a reframe: neuroimaging data, collected routinely in clinical contexts for BCI development, represents precisely the kind of organically produced, large-scale dataset that robotics currently lacks. Rather than constructing a new data pipeline, this project demonstrates how an existing one can be repurposed.

Expert intelligence

While the specific terminology for the expert varies among different works teleoperator, human operator, demonstrator for the scope of this document we define an expert as a human trained and tasked with producing high-quality demonstration data for a robotics learning pipeline. Attempts to collect data at scale for even large corporations such as Google require timelines spanning months and substantial investments in both time and monetary capital [4]. A new operator requires between 4 and 8 hours of training before their demonstrations are usable for policy training, and 20 to 40 hours before they produce consistently high-quality episodes. Quality filtering further compounds this; a 20 to 30 percent discard rate on collected demonstrations pushes the true cost of 500 usable demonstrations to \$800–\$1,200 when operator labor and senior engineering oversight are accounted for. In academic settings this cost is rendered invisible by graduate student labor, but the underlying expenditure of researcher time remains real regardless of how it is accounted for [5].

Systems Implication

Towards a Cross-Disciplinary Data Interface

Brain-computer interface engineering and robotics are both independently mature disciplines. Each has its own validated toolchains, standardised datasets, peer-reviewed classification frameworks, and active research communities. Yet despite this parallel maturity, the two fields remain architecturally siloed BCI pipelines are designed to produce decoded neural states, and robotics pipelines are designed to consume action labels, but no standardised interface layer exists to connect them. The data formats are incompatible, the timing constraints are different, and the validation requirements of each field were developed in isolation from the other. This is not a scientific problem. It is a systems integration problem.

Simproprioception demonstrates that this interface is technically feasible. A decoded EEG motor intent signal can be translated into a kinematic command, passed through a physics simulation, and produce stable, coherent robotic actuation using only components that already exist within their respective fields. The contribution is not a new algorithm. It is a proof that the transmission layer can be built, which reframes the challenge from a research question into an engineering one: how do we formalise, standardise, and scale this interface between two mature but disconnected systems?

The case for doing so is clearest in domains where the conventional LfD pipeline is not merely inefficient but genuinely unavailable. In assistive robotics, patients with motor impairments, stroke survivors, ALS patients, individuals with spinal cord injuries cannot physically perform demonstrations. Motor imagery BCI is not an alternative data source in this context; it is the only viable one. In hazardous industrial environments, the human operator dependency of teleoperation introduces risk that scales with task danger: deep-sea maintenance, nuclear facility operations, and high-voltage infrastructure are domains where removing the physical human from the control loop is not a convenience but a safety requirement. In both cases, a mature BCI-to-robotics interface would not just improve existing workflows, it would unlock applications that are currently out of reach.

The longer-horizon implication of this architecture is that motor intent is not the ceiling. The same pipeline that decodes imagined hand movement from sensorimotor rhythms is, in principle, extensible to higher-order cognitive states spatial attention, target selection, task sequencing. As BCI decoding methods mature and datasets scale, the resolution of what can be extracted from non-invasive neural signals will increase. A robotics system trained not on physical demonstrations but on cognitive intent would represent a fundamentally different class of human-machine collaboration one where the operator contributes decision-making rather than physical execution. This paper does not claim to build that system. It claims to demonstrate the first necessary component of it.

Technical Overview

Artifact Rejection via Independent Component Analysis (ICA)

The necessity of ICA for artifact attenuation in EEG pipelines is rooted in the physical constraints of the skull's volume conduction, which dictate that cortical and artifactual potentials mix linearly and instantaneously across the scalp — satisfying the mathematical prerequisites for Blind Source Separation [6]. Building on this, Jung et al. demonstrated that decomposing dense-array EEG into statistically independent subcomponents allows for the targeted subtraction of high-amplitude electrooculographic (EOG) and electromyographic (EMG) transients without attenuating the underlying sensorimotor rhythms [7]. Despite the proliferation of alternative filtering architectures, comprehensive evaluations of modern BCI topologies confirm that ICA remains the preeminent standard for isolating physiological noise while preserving neural signal fidelity [8].

Spatial Filtering via Common Average Referencing (CAR)

Following artifact decomposition, the pipeline employs Common Average Referencing (CAR) as an unsupervised spatial filter to enhance signal-to-noise ratio prior to covariance estimation. Raw EEG signals are fundamentally contaminated by global common-mode noise — residual environmental interference, reference-electrode bias, and distributed physiological drift. CAR mitigates this by computing the instantaneous arithmetic mean across all electrodes and subtracting it from each individual channel. The efficacy of this approach relies on the anatomically localised nature of Event-Related Desynchronization (ERD) in the sensorimotor bands during motor imagery — concentrated over the primary motor cortex beneath C3, C4, and Cz. As established by McFarland et al., the strictly localised nature of these motor potentials means their variance contributes negligibly to the global spatial average, ensuring CAR attenuates diffuse common-mode noise without distorting the localised target rhythms necessary for downstream decoding [9].

Feature Extraction: Riemannian Geometry over Common Spatial Patterns

The translation of EEG recordings into a feature space suitable for classification has traditionally relied on the Common Spatial Pattern (CSP) algorithm. However, CSP is fundamentally constrained by its eigenvalue decomposition mechanics: in a four-class motor imagery paradigm it requires complex one-versus-rest filter banks that are highly sensitive to intersession non-stationarity, prone to overfitting on limited calibration sets, and susceptible to localised channel noise. To overcome these structural limitations, this pipeline employs a Riemannian geometry framework. By estimating 22-channel EEG epochs as Symmetric Positive-Definite (SPD) covariance matrices, the spatial and temporal topology of the neural signal is preserved within its native non-Euclidean manifold. As demonstrated by Barachant et al., projecting these SPD matrices onto a Euclidean tangent space at the Riemannian mean effectively linearises the complex spatial boundaries of the signal [10]. Recent comprehensive literature reviews confirm that this tangent space mapping substantially outperforms multi-class CSP extensions in both robustness and computational efficiency[11].

Covariance Estimation via Ledoit-Wolf Shrinkage

Within the Riemannian framework, precise estimation of per-trial spatial covariance matrices is a strict prerequisite for stable tangent space projection. Relying on the standard maximum-likelihood Sample Covariance Matrix (SCM) introduces severe mathematical vulnerabilities when the number of temporal samples per epoch is small relative to channel dimensionality — causing sample eigenvalues to be systematically distorted, rendering matrices near-singular and degrading the Riemannian metric computation. To guarantee the stability of these SPD matrices, the pipeline applies Ledoit-Wolf shrinkage, which computes an analytically optimal shrinkage intensity to produce a convex linear combination of the empirical SCM and a perfectly conditioned target matrix [12]. Unlike heuristic regularisation techniques requiring expensive cross-validation, the Ledoit-Wolf estimator analytically minimises the expected Mean Squared Error between estimated and true covariance structures — definitively shown in neuro-engineering applications to outperform standard sample covariance and prevent spatial overfitting [13].

Classification via Linear SVM on Tangent Space Projections

Classification of EEG spatial covariance matrices traditionally necessitates complex non-linear architectures, introducing significant computational latency and overfitting risk under few-shot calibration constraints. This pipeline bypasses these limitations by coupling the Riemannian geometric transformation with a linear SVM. As formally established by Barachant et al., projecting SPD covariance matrices into a local Euclidean tangent space centred at the Riemannian geometric mean effectively unrolls and linearises the complex, non-linear class boundaries native to the EEG signal [14]. This tangent space mapping renders non-linear kernels computationally redundant — the linear SVM can draw optimal hyperplanes without kernel tricks, maximising real-time execution speed and preserving the mathematical stability required for physical hardware actuation.

Dynamic Calibration via Few-Shot Riemannian Transfer Learning

Traditional motor imagery pipelines mandate hundreds of preliminary trials per user to establish a functional classification boundary — inducing cognitive fatigue and degrading the target signal before functional actuation can begin. This pipeline implements a few-shot dynamic calibration framework via Riemannian transfer learning. Inter-subject non-stationarities driven by variations in skull geometry and cortical folding manifest as spatial shifts on the SPD manifold. As pioneered by Zanini et al., applying an affine transformation to align the Riemannian mean of a new target subject with the global population mean effectively normalises these inter-subject variances [15]. The pipeline then augments the population-wide baseline model with as few as 5–10 subject-specific trials per class, enabling dynamic fine-tuning of the SVM hyperplanes. Extensive literature on BCI transfer learning confirms that this targeted few-shot augmentation bridges the gap between zero-shot inefficiency and full calibration without subjecting the user to operational latency [16].

Command Smoothing via Majority-Vote Buffer

Single-epoch EEG decoding is inherently characterised by high trial-to-trial variance driven by momentary lapses in attention, transient physiological interference, or underlying BCI inefficiency. If instantaneous SVM outputs were mapped directly to the PyBullet physics engine as raw kinematic velocity vectors, the robotic end-effector would exhibit erratic high-frequency jitter, rendering smooth trajectory execution impossible. To bridge probabilistic neural decoding and deterministic robotic actuation, the pipeline implements a rolling majority-vote buffer that continuously aggregates the most recent k discrete predictions and requires strict statistical consensus prior to execution — acting as a temporal low-pass filter in the decision domain. As established in foundational literature on continuous brain-actuated control, temporal smoothing and evidence accumulation are strict necessities for physical BCI systems [17][18].

Robotic Simulation: KUKA iiwa in PyBullet

Decoded and smoothed motor intent commands are executed within a PyBullet physics simulation environment using a KUKA iiwa 7-DOF robotic arm loaded via its native URDF. Four motor imagery classes are mapped to four distinct joint velocity commands, providing a controlled and physically realistic testbed for validating the end-to-end pipeline without requiring physical hardware deployment.

Validation Consideration: BCI Illiteracy

Lower-bound performance observed in zero-shot transfer evaluation must be distinguished from algorithmic underfitting. Extensive neurophysiological literature establishes that high variance in motor imagery decoding is primarily driven by BCI Illiteracy — an estimated 15 to 30 percent of the population is fundamentally unable to produce the distinct sensorimotor rhythm modulations required for BCI control [19]. For these users, characteristic neural idle rhythms over the motor cortex either are anatomically absent or fail to reliably attenuate during imagined movement. Acknowledging this physiological bottleneck is essential: lower-bound outliers in classification metrics represent an expected biological reality rather than a structural failure of the Riemannian projection or SVM architecture.

Methodological Rigour: Preventing Data Leakage in Few-Shot Evaluation

Reported accuracy gains in few-shot calibration architectures are frequently scrutinised for potential overfitting and data leakage — which in EEG classification occurs when temporal correlations between adjacent epochs are inadvertently exploited during training. To systematically preclude these vulnerabilities, the pipeline enforces strict methodological constraints. As dictated by established guidelines for machine learning in neuroimaging [20][21], the evaluation framework respects the chronological arrow of physical deployment: calibration trials are drawn exclusively from initial, chronologically sequential data, strictly separated from the evaluation set. Spatial covariance matrices and Riemannian metrics are computed using only historically available data at each time step, ensuring zero informational bleed from the test set. This chronological stratification guarantees that reported transfer accuracy gains reflect genuine cross-subject generalisability rather than artifactual algorithmic memorisation.

Conclusion

Simproprioception does not propose a production-ready robotic control system, nor does it claim to resolve the data acquisition problem in robotics at scale. It claims something more modest and more foundational:

that the problem can be moved.

Learning from Demonstration has long treated the expert demonstrator as an unavoidable dependency, a fixed cost embedded in the architecture of the field. This paper argues for an alternative centered around neuroimaging data. The data robotics needs does not have to be manufactured from scratch by trained operators in controlled environments. It can be sourced from a domain where it already exists, where the collection infrastructure is mature, where quality standards are established, and where the subject pool is orders of magnitude larger than any LfD dataset ever constructed.

Simproprioception demonstrates that EEG motor imagery signals decoded through a Riemannian geometry pipeline and translated through a majority-vote buffer can produce stable, coherent commands for a simulated robotic arm. The pipeline is end-to-end, built on validated components, and reproducible.

The larger implication is that BCI and robotics, two fields that have matured in parallel without speaking to each other, now have a demonstrated basis for collaboration. Formalising the interface between them standardising the transmission layer, establishing shared validation frameworks, building the tooling that makes this pipeline replicable across research groups is the systems engineering work this project points toward.

From monkey see, monkey do. To monkey think, robots do. The shift is not just in the data. It is in what we believe the data can be.

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